

Design & Analysis of Algorithms [L-T-P:3-0-0 Credit: 3]

Prerequisites: Programming and Data Structures

Category: OLD (Modified)

Syllabus :

Introduction and basic concepts: Complexity measures, worst-case and average- case complexity functions, problem complexity, quick review of basic data structures and algorithm design principles. **[4 Lectures]**

Sorting and selection: Finding maximum and minimum, k largest elements in order; Sorting by selection, tournament and heap sort methods, lower bound for sorting, other sorting algorithms— radix sort, quick sort, merge sort; Selection of k-th largest element. **[8 Lectures]**

Searching and set manipulation: Searching in static table— binary search, path lengths in binary trees and applications, optimality of binary search in worst cast and average-case, binary search trees, construction of optimal weighted binary search trees; Searching in dynamic table -randomly grown binary search trees, AVL and (a, b) trees. **[6 Lectures]**

Hashing: Basic ingredients, analysis of hashing with chaining and with open addressing. Union-Find problem: Tree representation of a set, weighted union and path compression-analysis and applications. **[4 Lectures]**

Graph problems: Graph searching -BFS, DFS, shortest first search, topological sort; connected and biconnected components; minimum spanning trees— Kruskal's and Prim's algorithms— Johnson's implementation of Prim's algorithm using priority queue data structures. **[6 Lectures]**

Algebraic problems: Evaluation of polynomials with or without preprocessing. Winograd's and Strassen's matrix multiplication algorithms and applications to related problems, FFT, simple lower bound results. **[3 Lectures]**

String processing: String searching and Pattern matching, Knuth-Morris-Pratt algorithm and its analysis. **[3 Lectures]**

NP-completeness: Informal concepts of deterministic and nondeterministic algorithms, P and NP, NP-completeness, statement of Cook's theorem, some standard NP-complete problems, approximation algorithms. **[2 Lectures]**

References:

1. A. Aho, J. Hopcroft and J. Ullman; The Design and Analysis of Computer Algorithms, A. W. L, International Student Edition, Singapore, 1998

2. S. Baase: Computer Algorithms: Introduction to Design and Analysis, 2nd ed., AddisonWesley, California, 1988.
3. T. H. Cormen, C. E. Leiserson and R. L. Rivest: Introduction to Algorithms, Prentice Hall of India, New Delhi, 1998.
4. E. Horowitz and S. Sabni: Fundamental of Computer Algorithms, Galgotia Pub. /Pitman, New Delhi/London, 1987/1978.
5. K. Mehlhom: Data Structures and Algorithms, Vol. 1 and Vol. 2, Springer- Verlag, Berlin, 1984.
6. A. Borodin and I. Munro: The Computational Complexity of Algebraic and Numeric Problems, American Elsevier, New York, 1975.
7. D. E. Knuth: The Art of Computer Programming, Vol. 1, Vol. 2 and Vol. 3. Vol. 1, 2nd ed., Narosa/Addison-Wesley, New Delhi/London, 1973; Vol. 2: 2nd ed., Addison-Wesley, 18 London, 1981; Vol. 3: Addison-Wesley, London, 1973.
8. S. Winograd: The Arithmetic Complexity of Computation, SIAM, New York, 1980.

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